Android Developer Fundamentals

Hello World

Lesson 1



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1.2 Create Your First Android App

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Contents

- Android Studio
- Creating "Hello World" app in Android Studio
- Basic app development workflow with Android Studio
- Running apps on virtual and physical devices

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Prerequisites

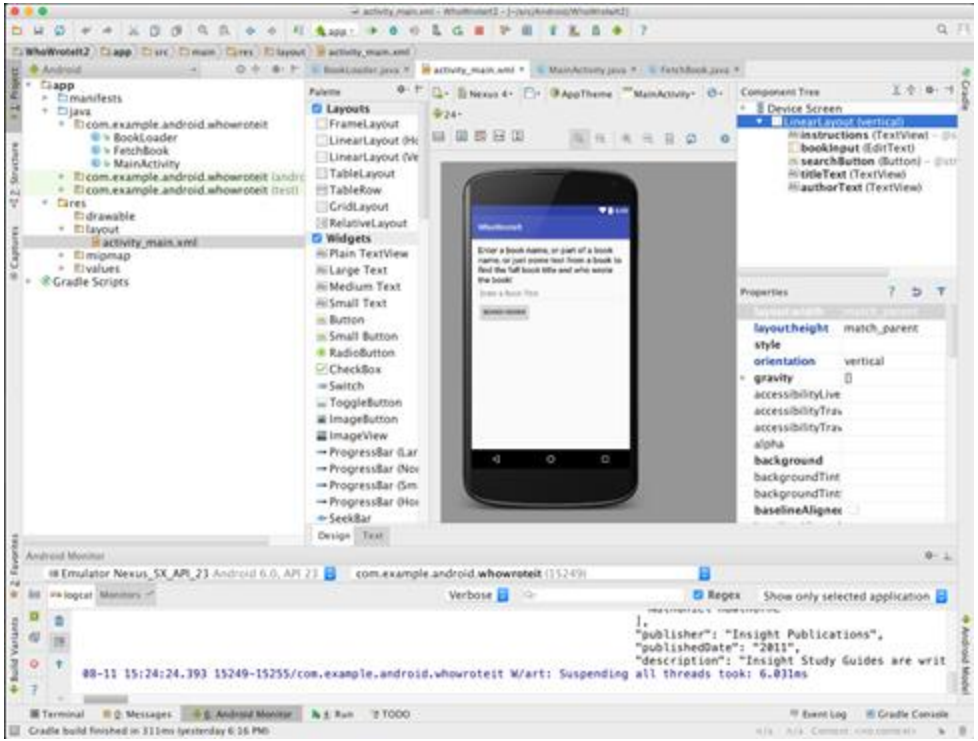
- Java Programming Language
- Object-oriented programming
- XML - properties / attributes
- Using an IDE for development and debugging

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Android Studio

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What is Android Studio?



- Android IDE
- Project structure
- Templates
- Layout Editor
- Testing tools
- Gradle-based build
- Log Console
- Debugger
- Monitors
- Emulators

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Installation Overview

- Mac, Windows, or Linux
- Requires Java Development Kit (JDK) 1.7 or better from Oracle Java SE downloads page
- Set JAVA_HOME to JDK installation location
- Download and install Android Studio from http://developer.android.com/sdk/index.html
- See 1.1 P Install Android Studio for details

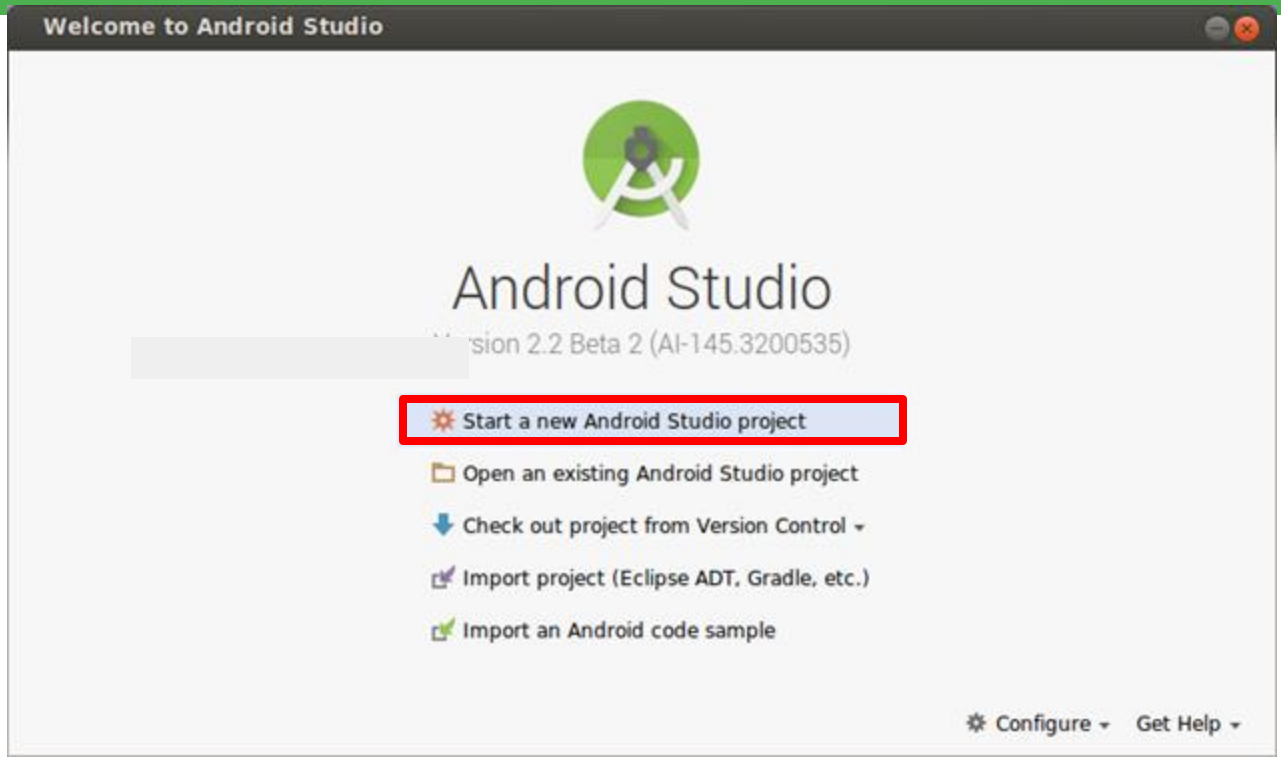
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Creating Your First Android App

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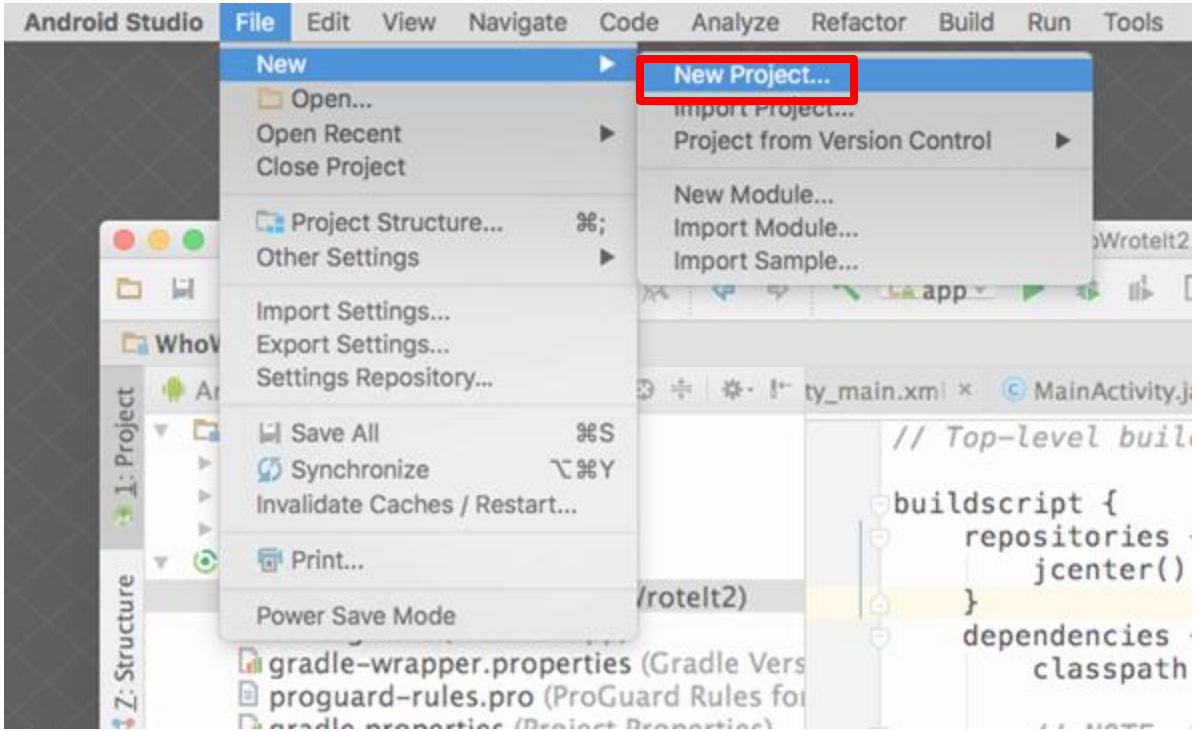
Start Android Studio





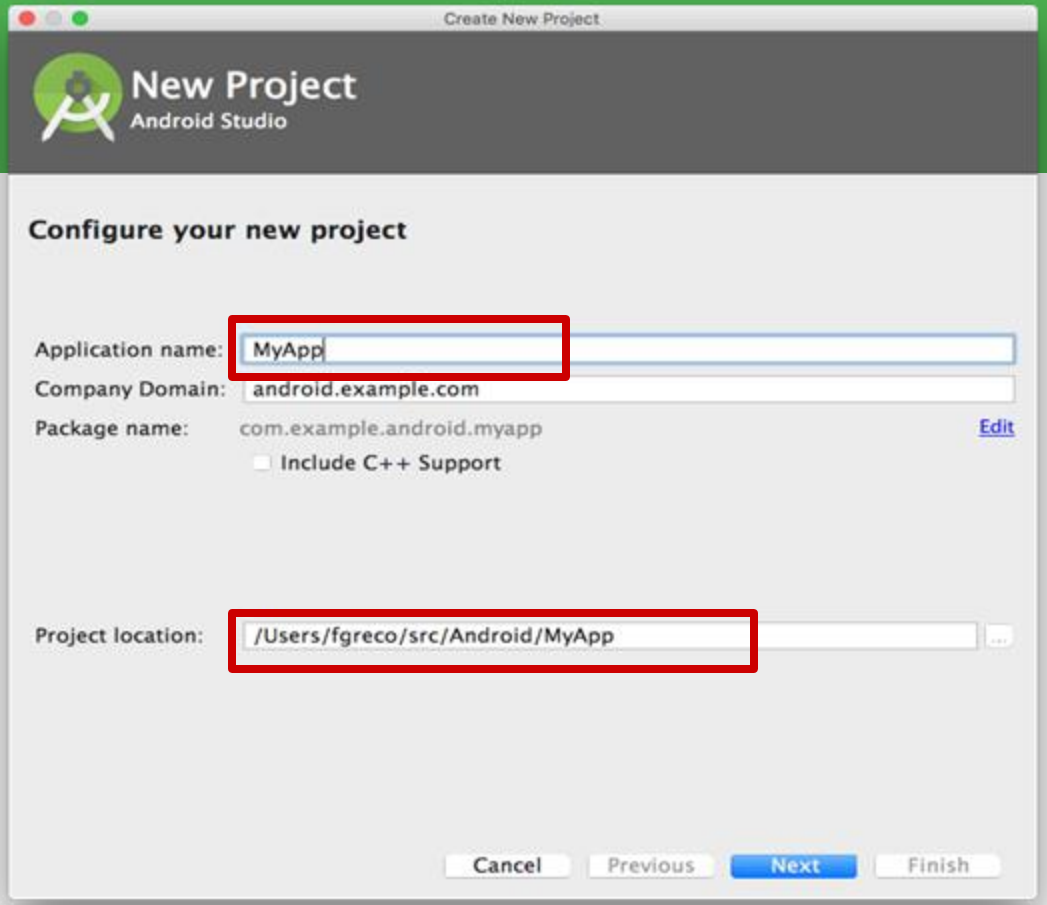
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Create a project inside Android Studio



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Name your app

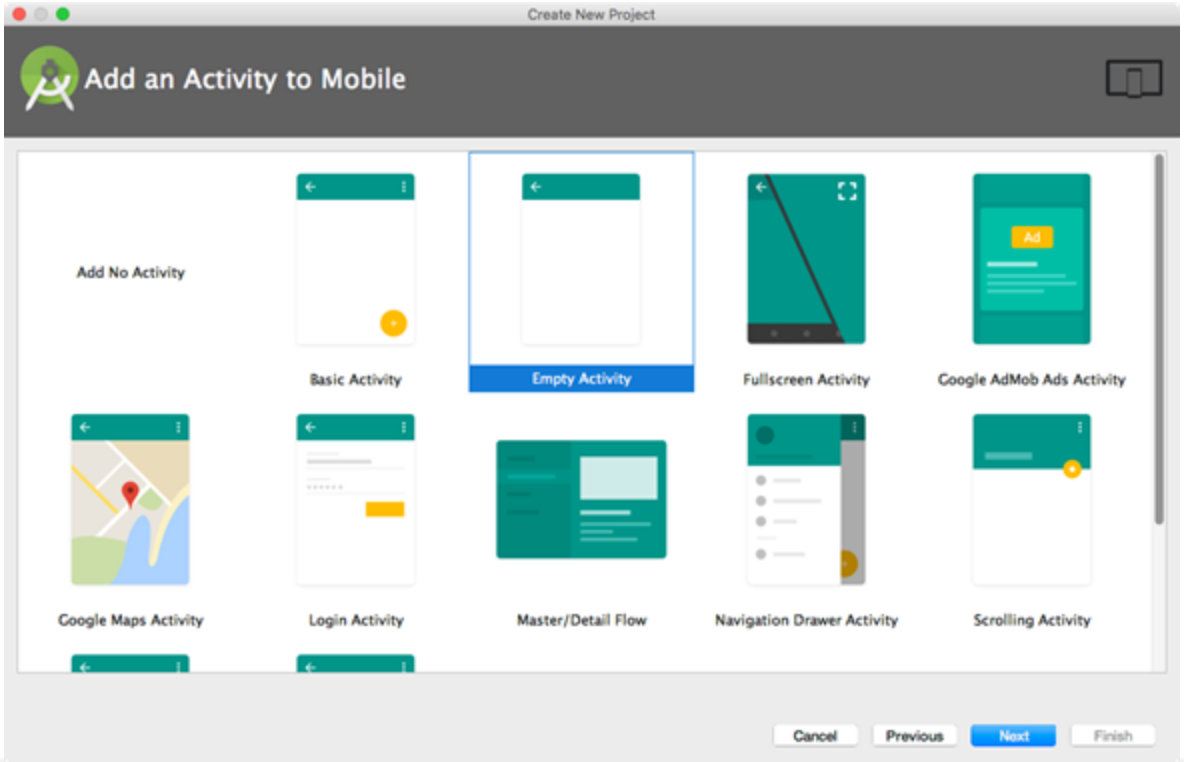


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Pick activity template

Choose templates for common activities, such as maps or navigation drawers.

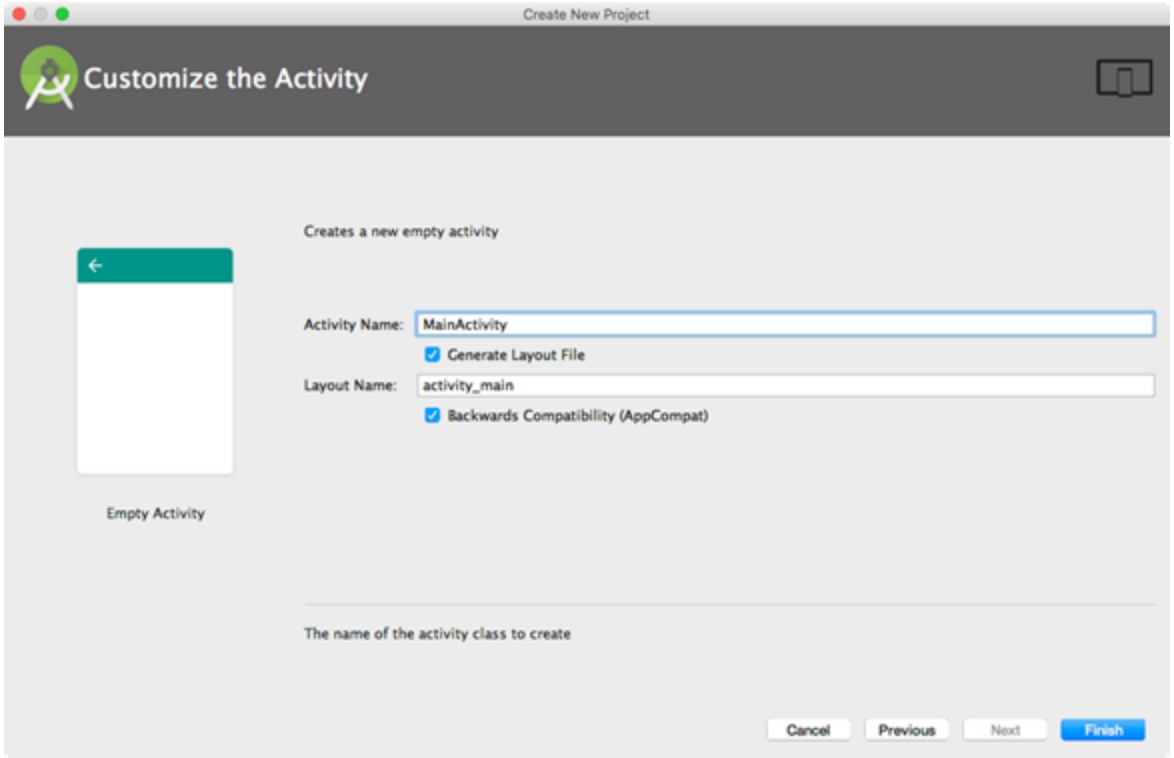
Pick Empty Activity or Basic Activity for simple and custom activities.



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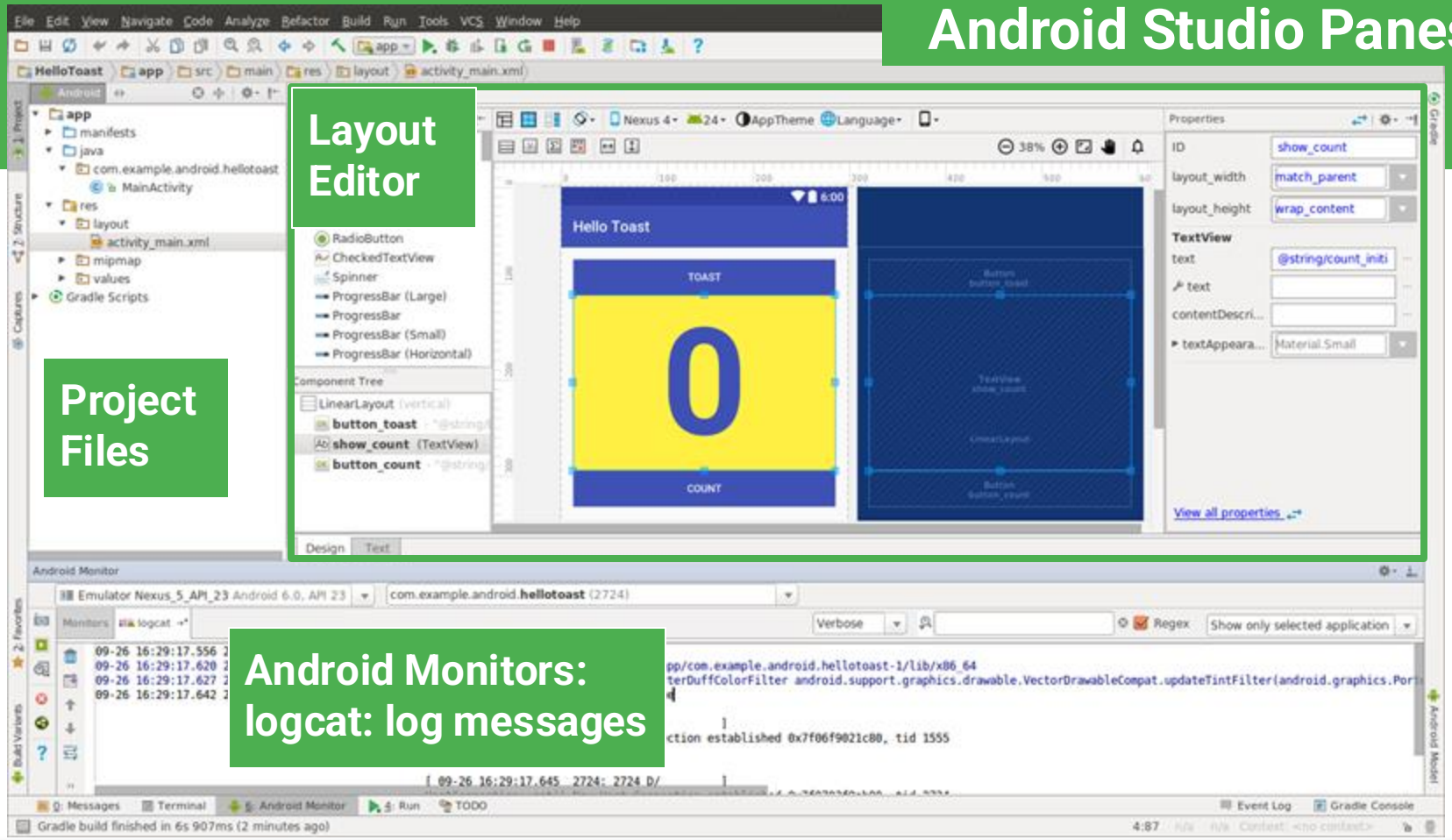
Name your activity

- Good practice to name main activity MainActivity and activity_main layout
- Use AppCompat
- Generating layout file is convenient



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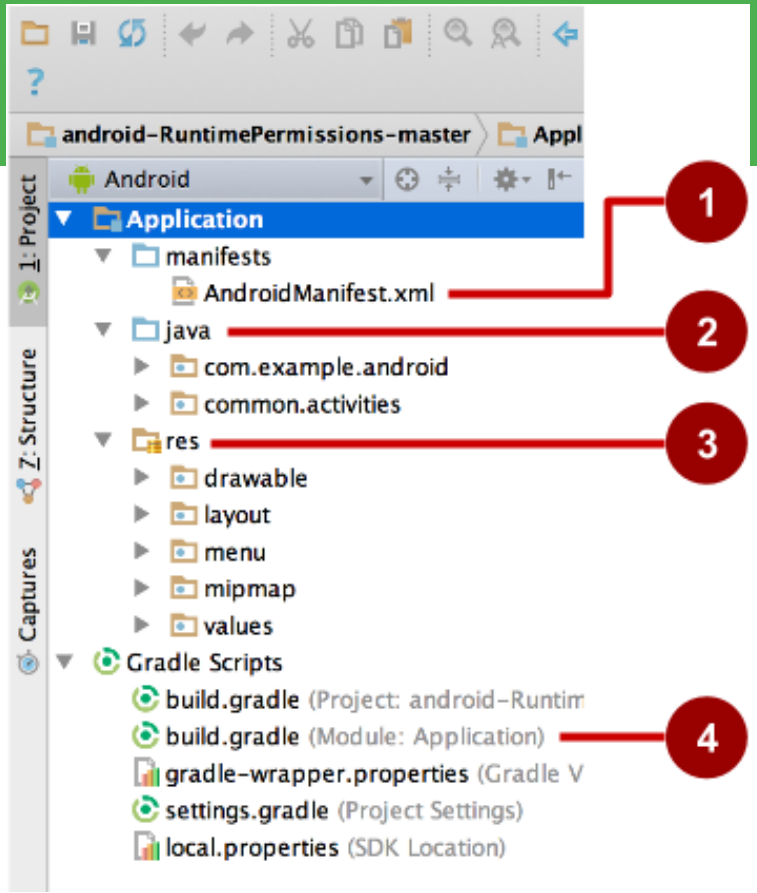
Android Studio Panes



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Project folders

1. **manifests**—Android Manifest file - description of app read by the Android runtime
2. **java**—Java source code packages
3. **res**—Resources (XML) - layout, strings, images, dimensions, colors...
4. **build.gradle**—Gradle build files



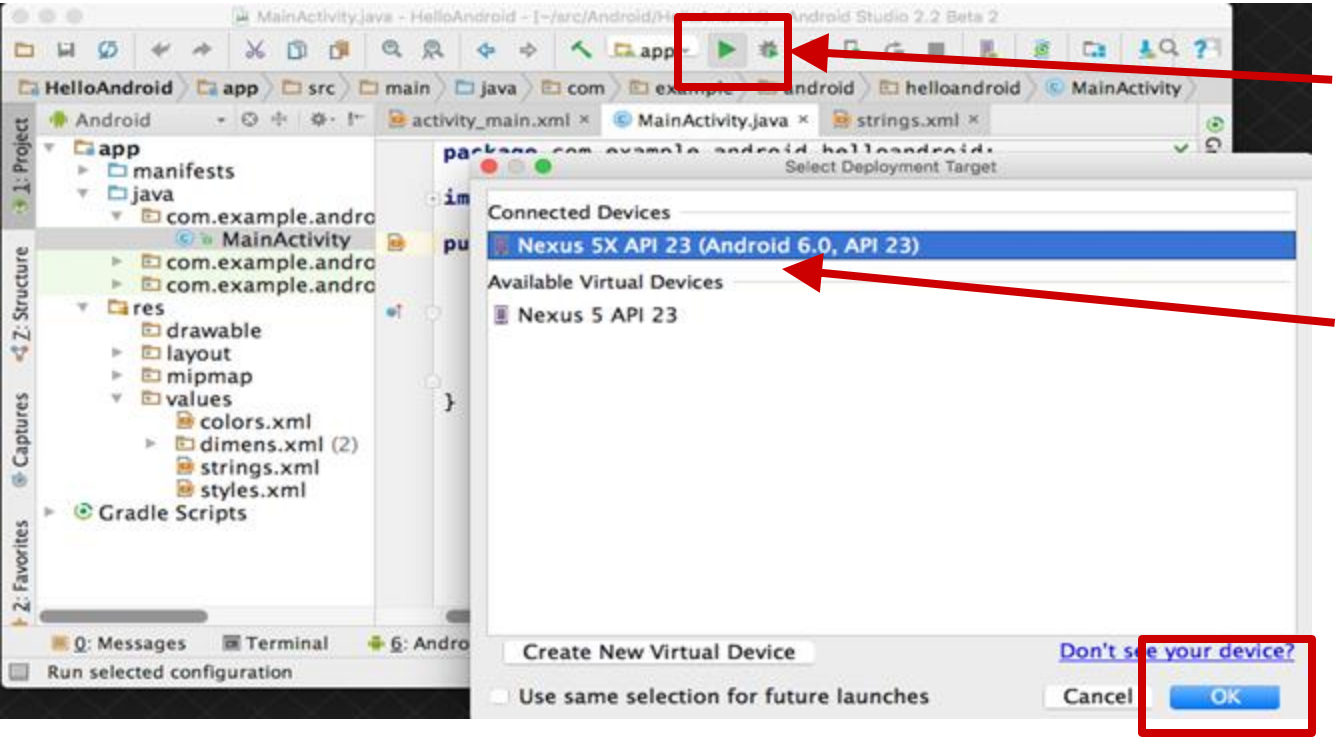
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Gradle build system

- Modern build subsystem in Android Studio
- Three build.gradle:
  - project
  - module
  - settings
- Typically not necessary to know low-level Gradle details
- Learn more about gradle at https://gradle.org/

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Run your app



1. Run
2. Select virtual or physical device
3. OK

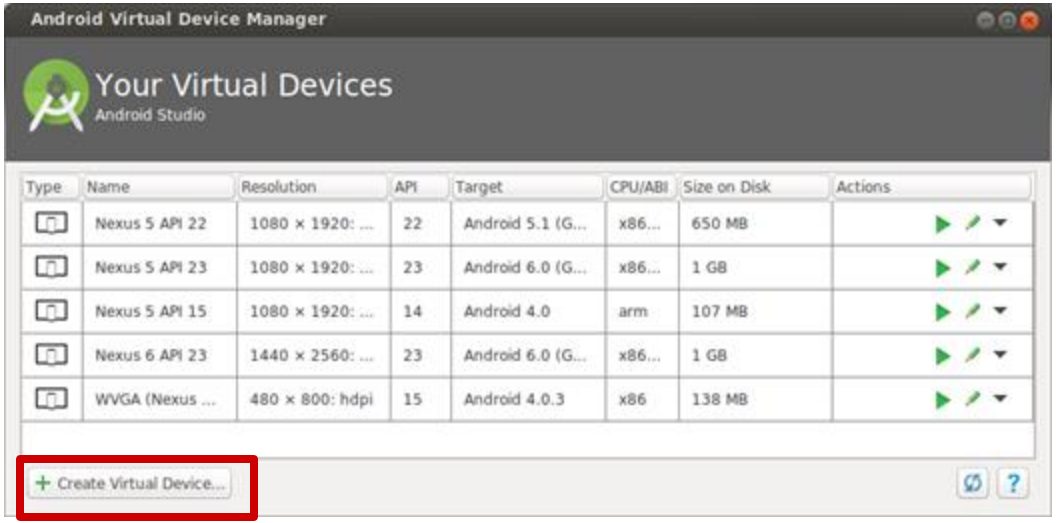
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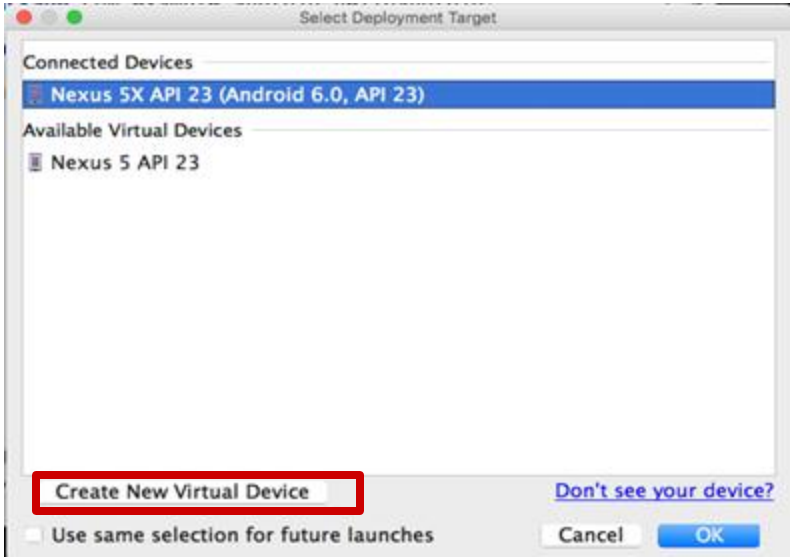
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Create a virtual device

Use emulators to test app on different versions of Android and form factors.

Tools > Android > AVD Manager or:

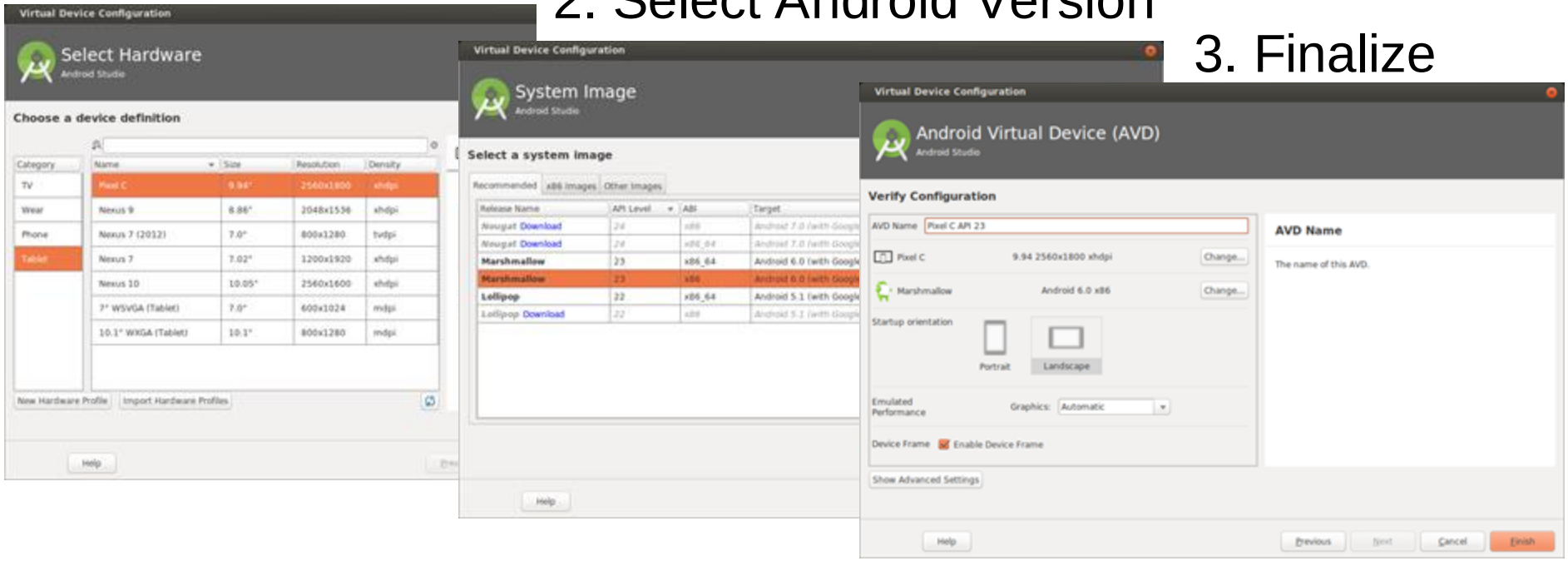




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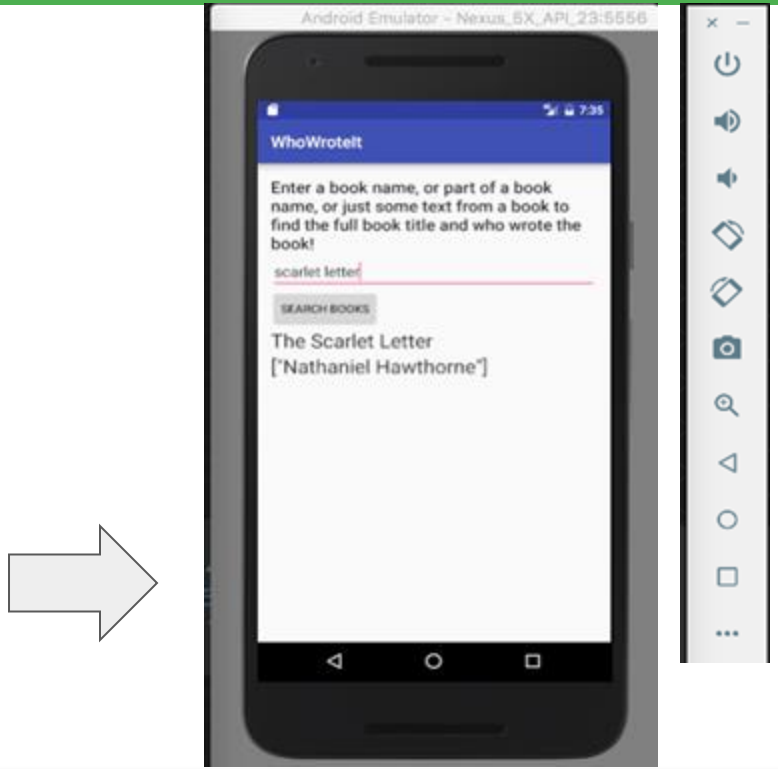
Configure virtual device

1. Choose hardware
2. Select Android Version
3. Finalize



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Run on a virtual device



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Run on a physical device

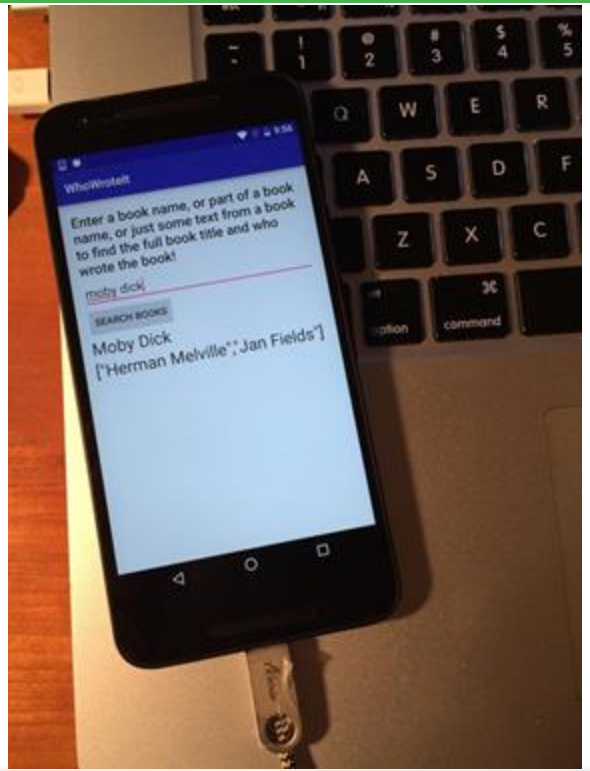
1. Turn on Developer Options:
   a. **Settings > About phone**
   b. Tap **Build number** seven times
2. Turn on USB Debugging
   a. **Settings > Developer Options > USB Debugging**
3. Connect phone to computer with cable

Windows/Linux additional setup:

- Using Hardware Devices

Windows drivers:

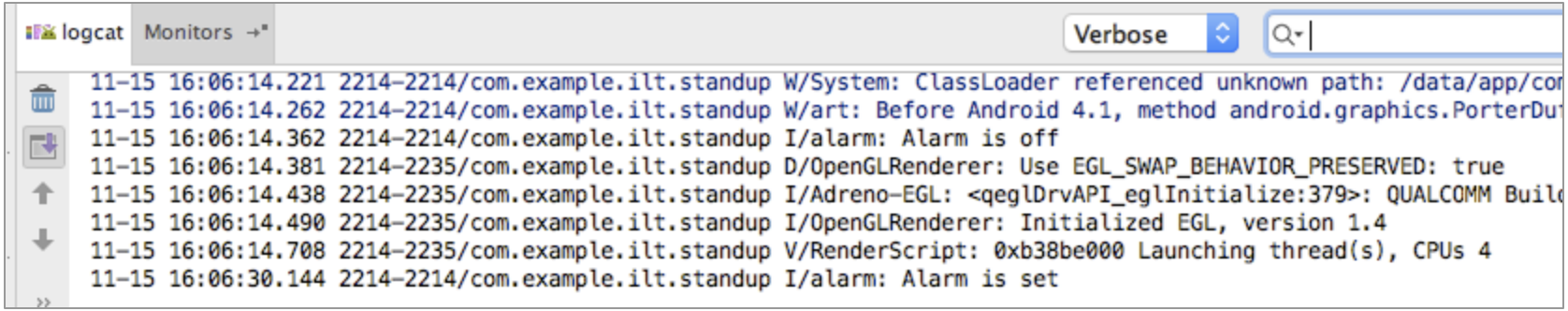
- OEM USB Drivers



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Get feedback as your app runs

- As the app runs, Android Monitor logcat shows information
- You can add logging statements to your app that will show up in logcat.



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Logging

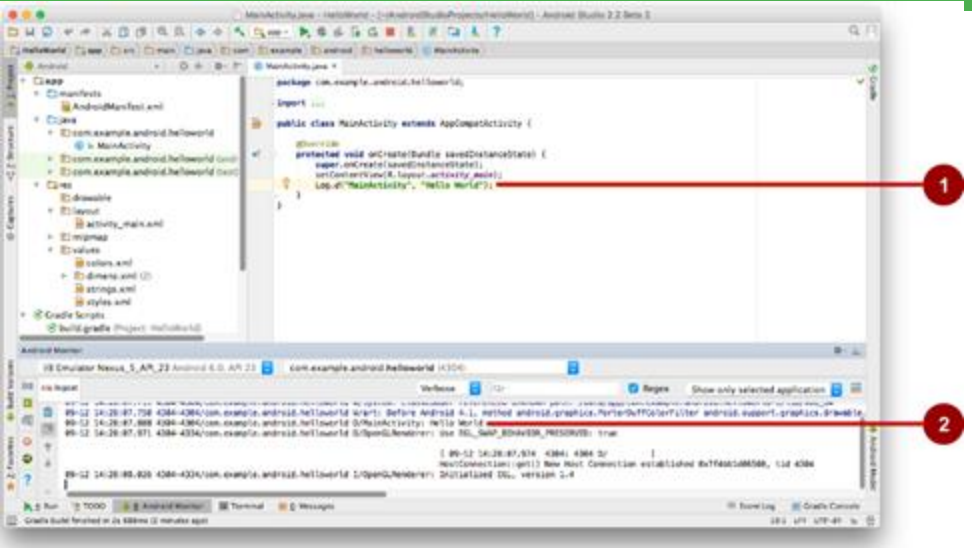
```
import android.util.Log;

// Use class name as tag
private static final String TAG =
    MainActivity.class.getSimpleName();

// Show message in Android Monitor, logcat pane
// Log.<log-level>(TAG, "Message");
Log.d(TAG, “Creating the URI…”);
```


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Android Monitor > logcat pane



1. Log statements in code.
2. logcat pane shows system and logging messages

- Set filters to see what's important to you
- Search using tags

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Learn more

- Meet Android Studio
- Official Android documentation at developer.android.com
- Create and Manage Virtual Devices
- Supporting Different Platform Versions
- Supporting Multiple Screens

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Learn even more

- Gradle Wikipedia page
- Google Java Programming Language style guide
- Find answers at Stackoverflow.com

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What's Next?

- Concept Chapter: 1.1 C Create Your First Android App
- Practical: 1.1 P Install Android Studio and Run Hello World

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